

Dave's Daily

AI • Bayesian optimization • US equities

tldr

1. White House asked OpenAI and Anthropic to hold new models from the UK AISI until a US security review; Anthropic already withheld Mythos 5.1.
2. The 10-year yield closed near 5.19%—highest since 2007—while equities stayed near highs; JPMorgan’s inverted-U frames when multiples de-rate.
3. BO for scaling-law construction (PLES / amortizing grids) meets regime-robust BO and Bayesian filters in equity and HFT market-making stacks.

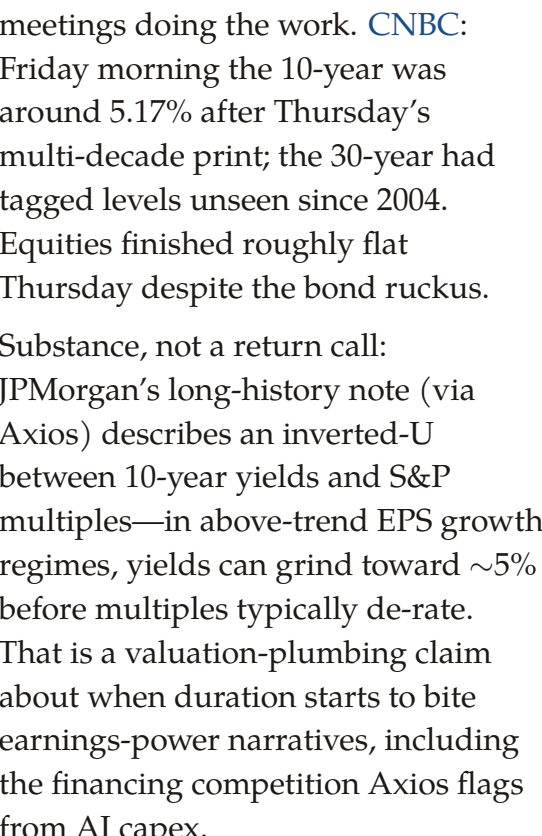
AI

[Reuters/CNA](#) / [TNW](#): Politico reported Thursday that the White House—via the Office of the National Cyber Director—asked OpenAI and Anthropic to withhold new frontier models from Britain’s [AI Security Institute](#) until US agencies finish their own review. A British official confirmed the report to Bloomberg on Friday. Anthropic has already limited *Claude Mythos 5.1* to US organizations; AISI’s Henry de Zoete noted the gap while saying the institute still tested OpenAI’s GPT-6 Astra pre-release.

The society story is alliance friction under capability pressure. Altman and Amodeli spent Wednesday at the UN Security Council urging common standards and verification; Washington is asserting a US-first gate on the same models. After the Australia Medicare agent incident disclosed this week, evaluation access looks less like a courtesy and more like a contested security asset—who gets to probe frontier systems before release, and on whose clock.

Algorithms / frontier. At The Information’s AI Agenda Live (23–24 Sep), Google DeepMind’s Koray Kavukcuoglu said [Gemini 4](#) has entered early post-training, targeting a ship “much earlier” than year-end after a July pre-train start—a compressed frontier cycle while evaluation politics tighten above.

Software / business. Enterprise agent governance keeps shipping beside the politics: [Dataiku](#) (24 Sep) launched cross-platform Agent Management plus Cobuild Insights/Headless; [UiPath Cartographer](#) (23 Sep) builds a governed “Map of Work” for coding agents.



Frontier training compute (running max of published estimates, log₁₀ FLOP). Source: [Our World in Data](#) / [Epoch AI](#) notable-models dataset. Capability climbs while access to evaluate those capabilities becomes a geopolitical lever.

Markets

The mechanism is a *discount-rate* regime, not a scoreboard. [Axios](#): the 10-year closed near 5.19% Thursday—highest closing level since July 2007—with energy shocks and hike odds for the next two FOMC meetings doing the work. [CNBC](#): Friday morning the 10-year was around 5.17% after Thursday’s multi-decade print; the 30-year had tagged levels unseen since 2004. Equities finished roughly flat Thursday despite the bond ruckus. Substance, not a return call: JPMorgan’s long-history note (via Axios) describes an inverted-U between 10-year yields and S&P multiples—in above-trend EPS growth regimes, yields can grind toward ~5% before multiples typically de-rate. That is a valuation-plumbing claim about when duration starts to bite earnings-power narratives, including the financing competition Axios flags from AI capex.

AI & BO → trading (incl. HFT)
[Cross-regime Bayesian optimisation](#) for tabular deep-learning equity signals (arXiv:2608.27076) selects hyperparameters against trading performance across statistically distinct market regimes—BO as a robustness tool, not a forecast. On the HFT side, [RL market making under regime-switching order flow](#) (arXiv:2609.11614) restores profitability by adding a Bayesian online change-point filter on directional flow bias plus queue-adjusted quote exposure—belief-state microstructure, not a price target. Adaptive-schedule BO for expensive portfolio oracles ([TPE-AS](#)) remains the sample-efficiency bridge between LLM HPO and trading engines.

Key numbers (this edition)	
Item	Figure
10-year close (Thu)	~5.19%
Prior comparable 10y close	Jul 2007
30-year intraday (Thu)	~5.50%
S&P 500 Thu move	≈flat
Mythos 5.1 → UK AISI	withheld
Gemini 4 phase	post-training
PLES vs grid budget	<10%
Amortizing vs full grid	10–100×

Bayesian optimization

[Power-Law Entropy Search \(PLES\)](#) remains the sharpest “BO meets LLM scaling” result: cost-aware multi-fidelity acquisition that reduces uncertainty in *optimal hyperparameter scaling laws* (learning rate and batch size vs model/data scale) rather than chasing a single black-box peak—accurate laws at under one-tenth grid-search cost.

[Amortizing Scaling Law Construction Costs](#) (4 Sep) casts collecting the compute–loss frontier itself as Bayesian optimization, then fills the dense grid with surrogate-fantasized evaluations—progressive budget expansion plus fantasization to match full-grid scaling-law fits at far lower train cost. Data-efficiency theme continues in multi-fidelity multi-scale mixture search. The trading papers above reuse the same instinct: spend scarce evaluations where regime uncertainty is highest, whether the black box is a pre-train run or a market-making policy.